TRIPATHI STUDY ZONE
JEE & NEET
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CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 4

(With Solutions)

Duration : 3 Hours **Max. Mark : 720**

Instructions

- The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- The question paper contains three parts of Physics, Chemistry and Biology respectively.
- Part A contains 45 Questions from Physics section. Part B contains 45 Questions from Chemistry section. Part C contains 90 Questions from Biology.

PART A PHYSICS

1. A satellite is launched into a circular orbit of radius R around the earth. A second satellite is launched into an orbit of radius 1.01 R. The period of the second satellite is larger than that of the first one by

(a) 0.5% (b) 1%
(c) 1.5% (d) 3%

2. A point initially at rest moves along X-axis. Its acceleration varies with time as $a = (3t - 2)\ \text{ms}^{-2}$. If it starts from origin, then the distance covered in 4s is

(a) 12 m (b) 20 m
(c) 15 m (d) 28 m

3. If a wire is stretched to make it 0.2% longer, its resistance will be

(a) increase by 0.4% (b) decrease by 0.4%
(c) increase by 0.1% (d) decrease by 0.1%

4. In a biprism experiment, fringes are obtained on the screen at a distance of 2m from the source, when a monochromatic light of wavelength 400 nm used. When, a convex lens is placed between the source and screen, images of the slits in two position 0.03 mm and 0.12 mm respectively, then fringewidth on the screen will be

(a) 0.0013 mm (b) 13.3 mm (c) 0.013 m (d) 1.33 mm

5. A black body at a temperature of 127°C radiates heat at the rate of 9 cal $\text{cm}^{-2}\ \text{s}^{-1}$. At a temperature of 527°C the rate of heat radiated per unit area in cal $\text{cm}^{-2}\ \text{s}^{-1}$ is

(a) 80 (b) 120 (c) 144 (d) 180

6. The respective speeds of five molecules are 1, 2, 3, 4 and 5 km/s. The ratio of their rms velocity and the average velocity will be

(a) $\sqrt{11} : 3$ (b) $3 : \sqrt{11}$ (c) 1 : 2 (d) 3 : 4

7. A coil in the shape of an equilateral triangle of side 0.02 m is suspended from its vertex, such that it is hanging in a vertical plane between the pole pieces of permanent magnet producing a uniform field of 5×10^{-2} T. If a current of 0.1 A is passed through the coil, what is the couple acting?

(a) $5\sqrt{3} \times 10^{-7}$ N-m (b) $5\sqrt{3} \times 10^{-10}$ N-m

(c) $\frac{\sqrt{3}}{5} \times 10^{-7}$ N-m (d) None of these

8. The radius of earth is 6400 km and $g = 9.8$ m/s^2. If the body placed at the equator has to become weightless the earth should make one complete rotation in

(a) 12 h (b) 1.4 h

(c) 6 h (d) 24 h

9. A zener diode having breakdown voltage equal to 15 V, is used in a voltage regulator. Circuit as shown in the figure, the current through the diode is

(a) 5 mA (b) 10 mA

(c) 15 mA (d) 20 mA

10. A pipe of length 85 cm is closed from one end. The number of possible natural oscillations of air column in the pipe whose frequencies lie below 1250 Hz are
(take, velocity of sound in air = 340 ms^{-1})

(a) 5 (b) 4

(c) 6 (d) 10

11. A uniform rod of length L and mass 2.0 kg is made to rest on two measuring scales at its two ends. A uniform block of mass 2.8 kg is placed on the rod at a distance of $\frac{L}{4}$ from its left end. The force experienced by the measuring scale on the right end is

(a) 10 N (b) 15 N

(c) 17 N (d) 19 N

12. Two pendulums of lengths 100 cm and 110.25 cm start oscillating in phase. After how many oscillations will they again be in same phase?

(a) 20 oscillations of smaller pendulum

(b) 21 oscillations of larger pendulum

(c) 21 oscillations of smaller pendulum

(d) None of these

13. An insulated box containing a diatomic gas of molar mass M moving with a speed v_0 is suddenly stopped.

The increase in gas temperature as a result of stopping the box is

(a) zero (b) $\frac{Mv_0^2}{2R}$ (c) $\frac{Mv_0^2}{3R}$ (d) $\frac{Mv_0^2}{5R}$

14. Work function of a metal is 5.2×10^{-18}J, then its threshold wavelength will be

(a) 736.7 Å (b) 760.7 Å (c) 301 Å (d) 344.4 Å

15. A lift of mass 1000 kg is supported by thick steel ropes. If maximum upward acceleration of the lift be 1.2 ms^{-2} and the breaking stress for the ropes be 1.4×10^8 Nm^{-2}, what should be the minimum diameter of rope?

(a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

16. Maximum efficiency of half-wave rectifier is

(a) 100% (b) 80% (c) 50% (d) 40%

17. The Rydberg constant for an electron revolving around hydrogen atom is R. The Rydberg constant for electron revolving about 10 times ionised sodium atom will be

(a) R (b) less than R

(c) greater than R (d) arbitrary

18. N small metal drops (identical) are charged to 10 V each. If they coalesce to form a single big drop, then its potential energy will be

(a) $10\,N^{1/3}$ (b) $\frac{10}{N^{2/3}}$ (c) $10\,N^{2/3}$ (d) $\frac{N^{2/3}}{10}$

19. Two glass capillary tubes of length L and $3L$ and radii R and $\frac{3R}{2}$ are connected in series. The net rate of flow of liquid through them will be (given, rate of the flow through single glass capillary, $Q = \frac{\pi p R^4}{8\eta L}$)

(a) 0.62 Q (b) 1.73 Q

(c) 1.6 Q (d) 1.38 Q

20. In an adiabatic change, the pressure and temperature of a monoatomic gas are related as $p \propto T^k$, where k equals

(a) $\frac{5}{2}$ (b) $\frac{2}{5}$ (c) $\frac{5}{3}$ (d) $\frac{3}{5}$

21. A source of emf 12 V and having negligible internal resistance connected to a variable resistor, so that current in the circuit changes with time as $\sqrt{i} = t - 1$. Then, total charge transfer by the source in first second will be

(a) 0.46 C (b) 0.33 C

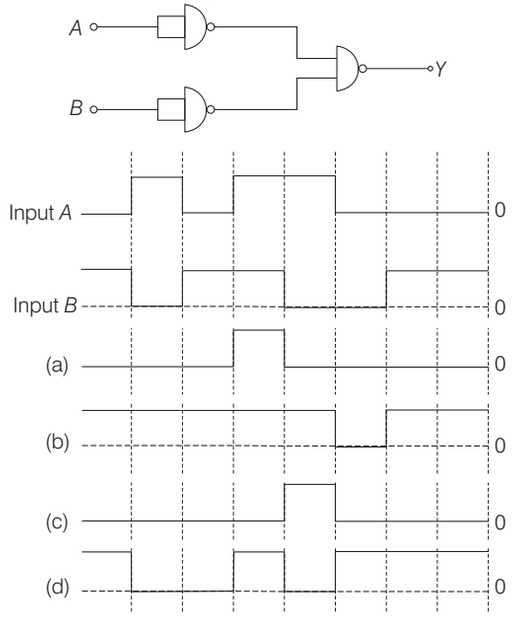
(c) 2.1 C (d) 1.2 C

22. A black hole is a body from whose surface nothing can escape. For a body of mass 9 times that of earth the radius for being a black hole is
(take, mass of earth = 6×10^{24} kg and
$G = 6.67 \times 10^{-11}$ Nm^2 kg^{-2})

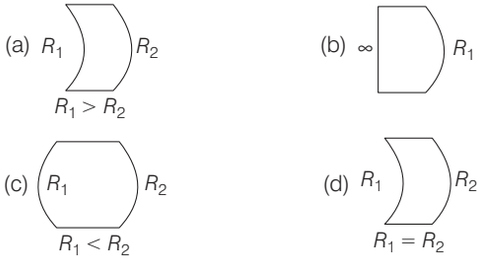
(a) 6 cm (b) 8 cm (c) 5 cm (d) 10 cm

23. A radioactive element ${}_{90}X^{238}$ decays into ${}_{83}Y^{222}$. The number of β -particles emitted are
(a) 1 (b) 2 (c) 4 (d) 6

24. The logic circuit shown below has the input waveforms A and B as shown in the figure. Pick out the correct output waveform.



25. In which of the following case of spherical lenses (as shown in the figure), the emergent ray is parallel to incident ray?



26. In a series L-C-R circuit with $L = 2$H, $C = 8\,\mu$F and $R = 5\,\Omega$, the Q-factor will be
(a) 50 (b) 100 (c) 150 (d) 200

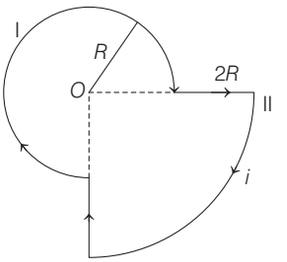
27. When a weight of 25 kg is suspended from a steel wire of length 5m and radius 0.2 mm. Its length increases by 2.5 cm. If the diameter of the wire is doubled, then the extension in its length will be
(a) 0.52 m (b) 0.62 cm (c) 0.75 cm (d) 0.83 cm

28. In a HCl molecule, the Cl can be treated on infinite mass and H oscillating alone. If the oscillation of HCl molecule shows frequency of 7×10^{13} s^{-1}, the value of force constant is
(take, Avogadro number = 6×10^{26} per kg-mole)
(a) 968 Nm^{-1} (b) 535.4 Nm^{-1} (c) 322.6 Nm^{-1} (d) 350 Nm^{-1}

29. A motor of power P_0 is used to deliver water at a certain rate through a given horizontal pipe. To increase the rate of flow of water through the same pipe n times, the power of the motor is increased to P_1. The ratio of P_1 to P_0 is
(a) n : 1 (b) n^2 : 1 (c) n^3 : 1 (d) n^4 : 1

30. A transparent cuboid contains a small air bubble, which apparent distance is 1.5 cm when seen through one face and 3.5 cm when seen through other face. If refractive index of transparent cuboid is 1.5, then real length of the edge of cube is
(a) 6 cm (b) 5 cm (c) 9 cm (d) 7.5 cm

31. Consider the circular loop having current i and with central point O. The magnetic field at the central point O is



(a) $\frac{2\mu_0 i}{3\pi R}$ acting downward (b) $\frac{5\mu_0 i}{12R}$ acting downward
(c) $\frac{6\mu_0 i}{11R}$ acting downward (d) $\frac{3\mu_0 i}{7R}$ acting upward

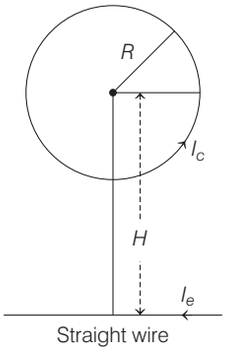
32. Two bodies of masses 2 kg and 5 kg are located at (1, –2) and (1, 3) respectively. Then, the centre of mass have coordinates at
(a) $\left(1, \frac{11}{7}\right)$ (b) $\left(\frac{11}{7}, 1\right)$ (c) $\left(\frac{1}{7}, \frac{11}{7}\right)$ (d) $\left(\frac{11}{7}, \frac{1}{7}\right)$

33. If E, M, L and G denote energy, mass, angular momentum and gravitational constant respectively, then the quantity $\left(\frac{EL^2}{M^5G^2}\right)$ has the dimensions of
(a) time (b) length (c) mass (d) angle

34. Two waves each of amplitude a and frequency f have a phase difference $\frac{\pi}{2}$. The amplitude and frequency of resultant wave due to their super position will be
(a) $\frac{a}{\sqrt{2}}, \frac{f}{2}$ (b) $\frac{a}{\sqrt{2}}, f$ (c) $2a, \frac{f}{2}$ (d) $\sqrt{2}a, f$

35. The temperature of a radiating body increases by 50%. Then, the increase in amount of radiation emitted will be approximately
(a) 185° (b) 385° (c) 406° (d) 607°

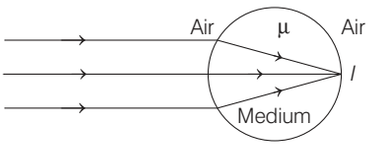
36. Circular loop of a wire and a long straight wire carry currents I_c and I_e, respectively as shown in the figure. Assuming that these are placed in the same plane. The magnetic field will be zero at the centre of the loop when the separation H is



(a) $\frac{I_e R}{I_c \pi}$ (b) $\frac{I_c R}{I_e \pi}$

(c) $\frac{\pi I_c}{I_e R}$ (d) $\frac{I_e \pi}{I_c R}$

37. In the given figure parallel rays of light incident on spherical medium of radius of curvature R meet at I. Then, refractive index of spherical medium is



(a) 2.5 (b) 1.5

(c) 1.35 (d) 2

38. If one of the slits in Young's double slit experiment is covered with a black opaque paper, then

(a) the fringe width decreases

(b) the bright fringes become fainter

(c) there will be uniform illumination all over the screen

(d) diffraction pattern will be observed

39. In a non-uniform electric field, the electric dipole experiences

(a) torque only (b) force only

(c) torque as well as net force (d) None of these

40. The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in weber, t is time in second and ϕ_0 is a constant, the output voltage across the secondary coil is

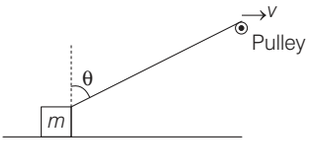
(a) 90 V (b) 120 V (c) 220 V (d) 30 V

41. A satellite is revolving around the earth, close to its surface with a kinetic energy E, then the energy required to be given to it, so that it escape from the earth is

(a) $2E$ (b) $\frac{E}{2}$

(c) E (d) $3E$

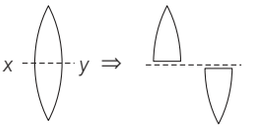
42. A block is dragged on a smooth plane with the help of a rope which moves with velocity v as shown in the figure. The horizontal velocity of the block is



(a) v (b) $v \sin\theta$

(c) $\frac{v}{\sin\theta}$ (d) $\frac{v}{\cos\theta}$

43. If a biconvex lens is cut into identical parts by a horizontal plane xy as shown in the figure, then which of the following statement is correct?



(a) Focal length of each part will be equal and forms image with same intensity.

(b) Focal length of each part will be same and forms image with reduced intensity.

(c) Focal length of each part will be different and forms image with reduced intensity.

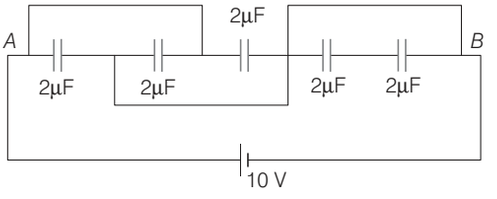
(d) None of the above

44. A mass m hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass m and radius R. Assuming pulley to be perfect uniform circular disc, the acceleration of the mass m, if the string does not slip on the pulley is

(a) $\frac{3}{2}g$ (b) $\frac{2}{3}g$

(c) $\frac{g}{3}$ (d) g

45. Total energy stored in the system given in figure is



(a) 3×10^{-4} J (b) 2.4×10^{-4} J

(c) 1.2×10^{-6} J (d) 2.4×10^{-6} J

PART B CHEMISTRY

46. The volume of 1.5 M HCl required to completely react with 2.4 g Mg is

$$Mg + 2HCl \longrightarrow MgCl_2 + H_2 \uparrow$$

(a) 103 mL (b) 133 mL
(c) 24.6 mL (d) 139 mL

47. If proton, electron and α-particle have equal kinetic energy, thier de-Broglie wavelengths would be in the order:

(a) $\lambda_e = \lambda_p < \lambda_a$ (b) $\lambda_a < \lambda_p < \lambda_e$
(c) $\lambda_a > \lambda_p < \lambda_e$ (d) $\lambda_p > \lambda_e < \lambda_a$

48. Among the following species, identify the isostructural pairs. [NF_3, NO_3^-, BF_3, $H_3\overset{+}{O}$, HN_3]

(a) [NF_3, NO_3^-] and [BF_3, H_3O^+]
(b) [NF_3, HN_3] and [NO_3^-, BF_3]
(c) [NF_3, H_3O^+] and [NO_3^-, BF_3]
(d) [NF_3, H_3O^+] and [HN_3, BF_3]

49. The strength of a solution for 6.82 g solute in 250 mL solution is

(a) 32.42 g/L (b) 25.62 g/L
(c) 27.28 g/L (d) 74.34 g/L

50. The density of a gas is 1.964 g dm^{-3} at 273 K and 76 mm Hg. The gas is

(a) CH_4 (b) CO_2
(c) Xe (d) C_2H_6

51. In an octahedral crystal field, the t_{2g} orbitals are

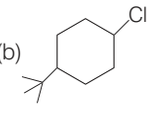
(a) raised in energy by 0.4 Δ_o (b) lowered in energy by 0.4 Δ_o
(c) raised in energy by 0.5 Δ_o (d) lowered in energy by 0.6 Δ_o

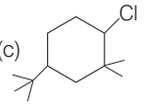
52. Which of the following is not the assumption of kinetic theory of gases?

(a) The actual volume of the gaseous molecules is negligible as compared to the total volume of the gas
(b) Molecules are perfectly elastic
(c) The critical temperature is the measure of the kinetic energy of the molecule
(d) The effect of gravity on motion of molecules is negligible

53. The most reactive substrate towards S_N2 reaction among the following is

(a) 

(b) 

(c) 

(d) 

54. The equilibrium constant for the reaction, $PCl_5(g) \longrightarrow PCl_3(g) + Cl_2(g)$ at 400 K if $\Delta H° = 77.2$ kJ mol^{-1} and $\Delta S° = 122$ JK^{-1} mol^{-1} is

(a) 1.958×10^{-4} (b) 2.931×10^{-6}
(c) 6.252×10^{-4} (d) 3.536×10^{-6}

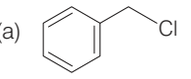
55. For an octahedral complex, which of the following *d*- electron configuration will give maximum CFSE?

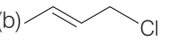
(a) High spin, d^6 (b) Low spin d^4
(c) Low spin, d^5 (d) High spin, d^7

56. If at 350K the concentration of [H_3O^+] is 10^{-6} M, then the value of K_w at this temperature will be

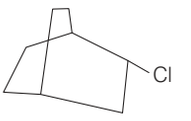
(a) 10^{-13} M^2 (b) 10^{-6} M^2
(c) 10^{-12} M^2 (d) 10^{-14} M^2

57. Which will not undergo S_N1 reaction?

(a) 

(b) 

(c) 

(d) 

58. The major product obtained on reaction of phenol with sodium hydroxide and CO_2 followed by acidification is

(a) pthalic acid (b) salicylic acid
(c) benzoic acid (d) salicylaldehyde

59. Which of the following species will acts as a Lewis acid but not as a Bronsted acid ?

(a) OH^- (b) O^{2-} (c) BF_3 (d) NH_2^-

60. Williamson's synthesis of preparing dimethyl ether is an/a example of

(a) free radical substitution (b) S_N2 mechanism
(c) S_N1 mechanism (d) electrophilic substitution

61. Which one of the following polymer is obtained by condensation polymerisation?

(a) Buna-N (b) Teflon
(c) Phenol-formaldehyde (d) Polyethene

62. Which of the following concentration term(s) represents solubility of BaF_2 in a solution of $Ba(NO_3)_2$?

(a) 2[NO_3^-] (b) $\frac{1}{2}[F^-]$
(c) [Ba^{2+}] (d) [F^-]

63. Value of coefficient p, q, r, s and t in the following redox reaction is

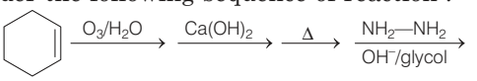
$$p\,Br_2 + q\,OH^- \longrightarrow r\,Br^- + s\,BrO_3^- + t\,H_2O$$

	p	q	r	s	t		p	q	r	s	t
(a)	3	6	1	5	3	(b)	3	6	5	3	1
(c)	3	6	5	1	3	(d)	3	5	1	6	3

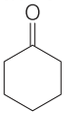
64. Which of the following statement(s) is/are incorrect?

(a) Argentite and cuprite are oxides
(b) Calamine and siderite are carbonates
(c) Malachite and chalcopyrite are the copper oxides
(d) Zinc blende and pyrites are sulphides

65. Consider the following sequence of reaction :



The end product of the given reaction is

(a) 
(b) 
(c) 
(d) 

66. 1 mole of FeC_2O_4 is oxidised by x mole of $Cr_2O_7^{2-}$ in acidic medium. Here, x is

(a) 3 (b) 1.5
(c) 0.5 (d) 1.0

67. The compound which cannot be stored in glass vessels is

(a) XeF_2 (b) XeO_3
(c) XeF_4 (d) XeF_6

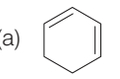
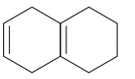
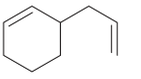
68. Nucleophilic reagents are the species that are negatively charged or have an electron rich atom. They are called Lewis base. Which of the following has highest nucleophilicity ?

(a) F^- (b) OH^-
(c) CH_3^- (d) NH_2^-

69. Which of the following does not react with dil. H_2SO_4 to give H_2O_2?

(a) SrO_2 (b) Na_2O_2
(c) BaO_2 (d) PbO_2

70. Conjugated dienes are more stable than non-conjugated dienes (both isolated and comulated) due to factors such as delocalisation of charge through resonance and hybridisation energy. Which of the following compound is most stable ?

(a) 
(b) 
(c) 
(d) 

71. Which of the following species is the leaving group in the alkaline hydrolysis of $CH_3\overset{\overset{O}{\|}}{C}OC_2H_5$?

(a) $C_2H_5O^-$ (b) $C_2H_5^+$
(c) $CH_3\overset{\overset{O}{\|}}{C}O^-$ (d) CH_3^-

72. The similarity between Li of group 1 and Mg of group 2 is called a diagonal relationship . It arises because

(a) Li and Mg atoms have similar sizes
(b) Li and Mg ions have similar sizes
(c) Li and Mg have similar electronegativities
(d) All of the above

73. Four successive members of the first series of the transition metals are listed below. For which one of them, the standard potential $E^\circ_{M^{2+}/M}$ value has a positive sign?

(a) Co(Z = 27) (b) Ni(Z = 28)
(c) Cu(Z = 29) (d) Fe(Z = 26)

74. Which of the following reagent can be used to distinguish between propyne and propene ?

(a) $AgNO_3$ in NH_3 (b) Alk. $KMnO_4$
(c) Br_2 in CCl_4 (d) Conc. H_2SO_4

75. Which of the following statements is incorrect about sodium hydroxide ?

(a) It is a white, translucent crystalline solid
(b) It is highly soluble in water to give a strong alkaline solution which is bitter in taste, corrosive and soapy to touch
(c) Crystals of NaOH are not hygroscopic
(d) If reacts with CO_2 in the atmosphere to form Na_2CO_3

76. Consider the following statements and choose the incorrect one.

(a) Acetamide is reduced to ethylamine by $LiAlH_4$
(b) Methyl isocyanide is hydrolysed to methylamine
(c) Methyl cyanide is reduced to methylamine by $LiAlH_4$
(d) Homologated 1° amine is obtained in the process

77. Which of the following compound can be most easily sulphonated ?

(a) Chlorobenzene (b) Toluene
(c) Nitrobenzene (d) Benzene

78. The correct statement regarding cycloaddition reactions is

(a) [2 + 2] is thermally allowed and [4 + 2] is photopchemically allowed

(b) [2 + 2] is photochemically allowed and [4 + 2] is thermally allowed

(c) Both [2 + 2] and [4 + 2] are thermally allowed

(d) Both [2 + 2] and [4 + 2] are photochemically allowed

79. The general electronic configuration for second row transition series is

(a) [Xe] $5d^{1-10}$, $5s^{1-2}$ (b) [Ar] $5d^{1-10}$, $4s^{1-2}$

(c) [Kr] $4d^{1-10}$, $5s^{1-2}$ (d) [Ne] $3d^{1-10}$, $4s^2$

80. Boric acid is used in carom boards for smooth gliding of pawns because

(a) H_3BO_3 molecules are loosely chemically bonded and hence, it is soft

(b) Its low density maked it fluffy

(c) It can be powdered to a very small grain size.

(d) H-bonding in H_3BO_3 give it a layered structure.

81. Packing fraction of simple cubic crystal lattice is

(a) 38% (b) 74%

(c) 68% (d) 52.4%

82. $10Cl^-(aq) + 2MnO_4^-(aq) + 16H^+(aq) \rightleftharpoons 5Cl_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$. The value of E° for this reaction is 0.15 V. What is the value of the equilibrium constant (K) for this reaction ?

(a) 2.6×10^{25} (b) 4.9×10^{12}

(c) 1.2×10^5 (d) 3.4×10^2

83. The form of carbon which is used as black pigment in black ink and filler in automobile tyres is

(a) graphite (b) charcoal

(c) coke (d) carbon black

84. If the rate constant of a reaction with virus is $4.8 \times 10^{-4}\ s^{-1}$, then the time required for the virus to become 80% inactivated is

(a) 48 mins (b) 62 mins

(c) 39 mins (d) 56 mins

85. Which of the following statement is incorrect regarding micelle?

(a) Aggregation of particles is known as micelle

(b) Micelle formation occurs below CMC

(c) Micelle formation occurs above CMC

(d) Micelle formation occurs above Kraft temperature

86. Radioactive material A has decay constant '8λ' and material B has decay constant 'λ'. Initially they have same number of nuclei. After what time the ratio of number of nuclei of material B to that A will be $1/e$?

(a) $\frac{1}{8\lambda}$ (b) $\frac{1}{7\lambda}$

(c) $\frac{1}{9\lambda}$ (d) $\frac{1}{\lambda}$

87. Match the given compound to number of unpaired electrons

	Compound		Number of Unpaired Electrons
A.	$[Cr(H_2O)_6]^{3+}$	1.	3
B.	$[Fe(H_2O)_6]^{2+}$	2.	4
C.	$[Zn(H_2O)_6]^{2+}$	3.	0
D.	$[Fe(CN)_6]^{4-}$	4.	0

Codes

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	1	2
(c)	2	1	4	3
(d)	4	3	1	2

88. The products obtained when copper metal is reacted with HNO_3 is

(a) HNO_2 or N_2 (b) NO or NO_2

(c) NO_2 or N_2O_5 (d) NO or N_2O_5

89. Glucose on reaction with ammoniacal silver nitrate gives silver mirror because it has

(a) alcoholic silver nitrate (b) ester group

(c) aldehyde group (d) ketone group

90. Dehydration of alcohols to alkenes on heating in presence of conc. H_2SO_4 involves

(a) protonation of alcohol molecule

(b) formation of an ester

(c) formation of carbocation

(d) elimination of water

PART C BIOLOGY

91. Which type of evolution is shown by wings of butterfly and birds?
(a) Convergent (b) Divergent
(c) Adaptive radiation (d) Chemical evolution

92. Which among the following is not a RNA virus?
(a) Avian leukemia virus (b) Bacteriophage MS-2
(c) Adenoviruses (d) Influenza virus

93. Which of the following carbohydrates is an oligosaccharide?
(a) Glyceraldehyde (b) Raffinose
(c) Dextrin (d) Inulin

94. Which among the following is not an effect of deforestation?
(a) Loss of biodiversity
(b) Disturbs the hydrologic cycle
(c) Soil erosion
(d) Genetic deformities

95. Which among the following statements is/are correct regarding velamen?
(a) Its cells comprise of fibrous thickenings
(b) It is highly hygroscopic
(c) Outermost tissue of the root
(d) All of the above

96. Which among the following is an incorrect statement?
(a) Runner has long internodes
(b) Stolon grows obliquely upwards and then bends down
(c) Offset develops from an axillary bud of an underground part of stem
(d) None of the above

97. An immunoglobin molecule contains two each of the heavy and light chains. Among these, the antigen binding site is present in
(a) constant region of both the light and heavy chains
(b) variable region of light chain
(c) variable region of heavy chain
(d) variable region of both the light and heavy chains

98. The schizont stage of *Plasmodium* parasite in human occurs in
(a) liver cells
(b) spleen
(c) liver and erythrocytes
(d) liver, spleen and bone marrow

99. Which of the following is a function of boron?
(a) Sugar metabolism
(b) Nitrogen metabolism
(c) Activation of enzymes
(d) Acting as enzyme cofactor

100. In which of the following conversions does oxidative decarboxylation take place?
(a) Succinic acid → Fumaric acid
(b) Pyruvate → Acetyl Co-A
(c) Fumarate → Malate
(d) Isocitrate → Citrate

101. The substances like fructose and some amino acids are absorbed with the help of the carrier molecules. This mechanism is known as the
(a) simple diffusion (b) facilitated transport
(c) active transport (d) osmosis

102. *Monascus purpureus* is used to produce
(a) cyclosporin-A (b) statins
(c) riboflavin (d) L-ascorbic acid

103. In glycolysis, which of the following reactions is catalysed by enzyme aldolase?
(a) Fructose-6-phosphate → Fructose 1, 6- biphosphate
(b) Dihydroxyacetone phosphate → Glyceraldehyde 3-phosphate
(c) Fructose 1, 6-biphosphate → Glyceraldehyde 3-phosphate + Dihydroxyacetone phosphate
(d) 2-phosphoglycerate → Phosphoenol pyruvate

104. During photolysis of water, the oxygen evolving complex contains the following elemental ions.
(a) Mg^{2+}, Cl^- and Ca^{2+} (b) Mg^{2+}, K^+ and Cl^-
(c) Mn^{2+}, Cl^- and Ca^{2+} (d) Mn^{2+}, K^+ and Cl^-

105. What are pyrenoids associated with?
(a) Starch storage (b) Photosynthesis
(c) Fatty acid oxidation (d) Enzyme secretion

106. Apical cell theory states that
I. a single apical cell is the structural and functional unit of apical meristem.
II. it governs the whole process of primary growth.
III. found in lower plants like algae and bryophytes.
IV. histogens form vascular strand.

Choose the correct option.
(a) I and II (b) III and IV
(c) I, II and III (d) II, III and IV

107. Immunity is a state of protection from foreign molecules. It is of two types, i.e. innate and adaptive immunity. Specificity, diversity , memory are characteristics of
(a) innate immunity
(b) non-specific immunity
(c) acquired immunity
(d) external defence

108. Match the items given in Column I with those in Column II and select the correct option.

	Column I		Column II
A.	Root pressure	1.	Cohesion and adhesion of water molecules
B.	Capillary action	2.	Evaporation of water from leaf surface
C.	Transpiration pull	3.	Entry of water into xylem by osmosis

Codes

	A	B	C		A	B	C
(a)	1	3	2	(b)	1	2	3
(c)	3	2	1	(d)	3	1	2

109. The advantage of a DNA molecule having two complementary strands is that
(a) easy pairing between chromatids
(b) transcription and translation processes can occur simultaneously
(c) semiconservative replication
(d) None of the above

110. Which of the following is an enzyme which recognises specific base sequences in DNA and cut the DNA at that site?
(a) Restriction enzymes (b) DNA ligase
(c) Helicase (d) Primase

111. Which among the following statements is/are correct regarding gemmules?
I. Gemmules arise as modified branches.
II. In case of *Marchantia*, the gemmae are multicellular.
III. In case of algae, the gemmae occur in *Sphacelaria*.
IV. In bryophytes, the gemmae are approximately not present in Bryopsida.

Choose the correct option.
(a) I and II (b) III and IV (c) I, II and III (d) All of these

112. Which among the following is/are charcteristic of imbibition?
(a) Development of pressure (b) Production of heat
(c) Volume change (d) All of these

113. A mango merchant desires to obtain all the mangoes in his orchard to have the taste and quality of alphonso mango. He tries to achieve this by using the grafting technique. Which variety would be used for the scion and the stock, respectively, to get the desired result?
(a) Scion of desi mango on stock of alphonso mango
(b) Scion of alphonso mango on stock of desi mango
(c) It can be either way out depending on the age of the plants
(d) Scion of desi mango on stock of both desi and alphonso are combined

114. Hydatid cyst is the larvae of
(a) *Rickettsia* (b) *Taenia solium*
(c) *Phlebotomus sp.* (d) *Echinococcus*

115. Early diagnosis of diseases cannot be done by
(a) recombinant DNA technology
(b) polymerase chain reaction
(c) enzyme linked immuno sorbent assay
(d) urine analysis

116. The pollen tube contains vegetative nucleus and a generative nucleus which are identical because
(a) they are products of amitosis
(b) they are products of mitosis
(c) they are products of meiosis
(d) they are products of meiosis-I

117. Fertilizin is a compatibility receptor protein found in
(a) inner plasma membrane (b) zona pellucida
(c) corona radiata (d) ooplasm

118. Match Column I with Column II and choose the correct option from the codes given below.

	Column I (Type of stem)		Column II (Characteristic)
A.	Monocot stem	1.	Vascular bundle closed
B.	Dicot stem	2.	Vascular bundle open
C.	Pith in monocot stem	3.	Pith absent
D.	Pith in dicot stem	4.	pith made up of parenchymatous cells

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	2	1	3	4
(c)	4	1	3	2	(d)	4	3	1	2

119. Heterostyly is an outbreeding device in which,
(a) pollen grains are unable to germinate on the stigma of same flower
(b) pollen grains mature earlier than the receptive time of stigma`
(c) style and stamens in a flower are at different heights
(d) structure of pollen sac and stigma functions as hurdles

120. Select the incorrect statements.
I. Bulbils are fleshy buds produced in the axil of foliage leaves in place of axillary buds.
II. Segments of *Opuntia* and other cacti produce new plants after falling on the ground.
III. In *Bryophyllum*, marginal buds form plantlets while they remain attached to the plants.
IV. Fleshy adventitious roots take part in vegetative propagation in onion.

Choose the correct option.
(a) I and II (b) III and IV
(c) Only II (d) Only IV

121. Repeated selfing will lead to
(a) increased heterozygosity
(b) increased homozygosity
(c) production of hybrids
(d) Homozygosity in some and heterozygosity in other traits

122. In prophase-I of meiosis, during which substage can terminalisation be observed?
(a) Pachytene (b) Diplotene
(c) Diakinesis (d) Karyokinesis

123. Given below are different types of microspore tetrads with their examples. Select the option, which is matched incorrectly.
(a) Decussate – *Magnolia*
(b) T-shaped – *Crocus*
(c) Isobilateral – *Aristolochia elegans*
(d) Linear – *Halophia*

124. Choose the incorrect statement(s).
I. Dislocation involves a damage to the ligaments surrounding the joints.
II. Sprain is a twisting of joint without dislocating it .
III. Muscular dystrophy is an autoimmune disorder.
IV. Gout is an inherited disorder of purine metabolism.
(a) I and II (b) Only III
(c) III and IV (d) II and III

125. Which among the following statements is incorrect regarding gymnosperms?
(a) Seeds are naked
(b) Double fertilisation is absent
(c) Xylem consists of vessels and tracheids
(d) Phloem consists of sieve cells

126. Which of the following statements is/are correct for Abscisic Acid (ABA)?
(a) ABA is translocated by both xylem and phloem
(b) It is synthesised in mature old leaves
(c) It is synthesised from mevalonic acid
(d) All of the above

127. Identify the correctly matched pair.
I. Acoelom – There is no body cavity
II. Pseudocoelom – Coelom is not lined by mesoderm
III. Eucoelom – Body cavity develops entirely within the mesoderm
IV. Enterocoel – Deuterostomes are enterocoeloms
(a) I and II (b) III and IV
(c) II, III and IV (d) All of these

128. Starch is made up of two polymers.
(a) amylose, amlodipine (b) amylose, amylopectin
(c) glucose, fructose (d) glucose, sucrose

129. Match Column I with Column II and choose the correct option from the codes given below.

	Column I (Reproductive organ)		Column II (Characteristic)
A.	Protonema	1.	Vegetative propagation
B.	Antheridium	2.	Male sex organ
C.	Archegonium	3.	Female sex organ
D.	Sporophyll	4.	Bears sporangia

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	2	1	3	4
(c)	1	2	4	3	(d)	2	1	4	3

130. Creatine excretion is probably associated with
I. histolysis.
II. histogenesis.
III. seen in foetus and pregnant women.
IV. presence of benzoic acid in diet.
The correct option is/are
(a) I, II and III (b) III and IV (c) I and II (d) I and III

131. Identify the incorrectly matched pair.
(a) Dairy — *Bacterium lactici acidi*
(b) Vinegar industry — *Acetobacter aceti*
(c) Alcohol industry — *Clostridium acetobutylicum*
(d) Tobacco industry — *Mycococcus codisans*

132. Which among the following are the chracteristics of resemblance between gymnosperms and angiosperms?
I. Secondary growth is not present in roots.
II. Secondary wood may be monoxylic or polyxylic.
III. Presence of ovule.
IV. Presence of endosporic and reduced gametophytes.
Choose the correct option.
(a) I and II (b) II and III (c) III and IV (d) II, III and IV

133. Which of the following is not a hormone releasing Intra Uterine Device (IUD)?
(a) Progestasert (b) LNG-20
(c) Cervical cap (d) Mirena

134. Which among the following statements are correct regarding cyanobacteria?
(a) Possess prokaryotic type of cells
(b) Incipient nucleus
(c) Asexual reproduction takes place by fission and fragmentation
(d) All of the above

135. Which among the following is the criteria for an element to be considered as essential?

I. The element must be essential for normal growth and reproduction.
II. The requirement of element must be specific.
III. It cannot be replaced by any another element.
IV. The requirement must be direct that is, not the result of any indirect effect.

Choose the correct option.

(a) I and II (b) III and IV
(c) II and III (d) I, II, III and IV

136. The post-transcriptional modification involves

I. Methyl guanosine triphosphate, an unusual nucleotide is added to the 5' end of *hn*RNA.
II. Adenylate residues (200-300) are added at 3'-end in a template independent manner.
III. Processed *hn*RNA is now called *m*RNA.
IV. *m*RNA is transported out of the nucleus.

Choose the correct option.

(a) I and II (b) IV and III (c) II, III and IV (d) I, II III and IV

137. Human dentition is correctly described by which set of terminologies?

(a) Thecodont, diphyodont, homodont
(b) Pleurodont, acrodont, homodont
(c) Thecodont, diphyodont, heterodont
(d) Pleurodont, acrodont, heterodont

138. Which among the following is/are an evidence for natural selection?

(a) Rate of reproduction and competition in nature
(b) Production of new varieties of plants and animals
(c) Mimicry and protective colouration
(d) All of the above

139. *Contagium vivum fluidum* for viruses was given by

(a) Mayer (b) Ivanowski (c) Stanley (d) Beijerinck

140. The five types of histone proteins are

(a) H_1,H_2A,H_2B,H_3 and H_4 (b) H_1,H_2,H_3,H_4 and H_5
(c) H_1A,H_1B, H_2, H_3 and H_4 (d) H_1, H_2,H_3A,H_3B and H_4

141. In the apoplast pathway,

(a) water moves exclusively through the cell wall without crossing any membrane
(b) water travels from one cell to the next *via* the plasmodesmata
(c) water crosses at least two membranes
(d) transport across the tonoplast is also involved

142. During implantation, which of the following does not occur?

(a) blastocyst attaches to the uterine wall
(b) the trophoblast cells produce enzymes that allow the blastocyst to 'eat its way' into the thick endometrium
(c) by 12th to 20th day, the blastocyst is completely implanted
(d) None of the above

143. Which of the following is in the proper order for the vectors given below in terms of increasing cloning capacity?

(a) BAC, cosmid, phage, plasmid, YAC
(b) YAC, BAC, cosmid, phage, plasmid
(c) Plasmid, cosmid, phage, BAC, YAC
(d) Plasmid, phage, cosmid, BAC, YAC

144. The presence of HCl in the stomach helps the digestion process in which of the following ways?

(a) Kills microorganims of food
(b) Maintains the pH of the stomach
(c) Preserves the food when in stomach
(d) All of the above

145. Match the items in Column I (National parks / Biosphere reserves) with Column II (location) and select the correct option.

	Column I		Column II
A.	Keoladeo National Park	1.	Odisha
B.	Dachigam National Park	2.	Bharatpur
C.	Similipal	3.	Kashmir

Codes

	A	B	C		A	B	C
(a)	2	3	1	(b)	1	2	3
(c)	3	1	2	(d)	3	2	1

146. Choose the correct statements regarding sucrose.

I. It is a monosaccharide.
II. It is a reducing sugar.
III. It is a disaccharide made up of glucose and fructose.
IV. Glucose and fructose are joined together by α-1,2 glycosidic linkage.

(a) Only II (b) Both II and III
(c) Both III and IV (d) None of these

147. Okazaki fragments are

(a) lagging strand
(b) each Okazaki fragment has its own RNA primer
(c) replicated in a discontinuous manner
(d) All of the above

148. Arrange the following processes of biotic succession in a sequence.

I. Nudation II. Migration
III. Ecesis IV. Aggregation
V. Competition VI. Reaction
VII. Stabilisation

Choose the correct option.

(a) I → II→ III → IV → V → VI → VII
(b) II → I → III → IV → V → VI →VII
(c) II → I → IV → III → V→ VI → VII
(d) I → II → IV → III → V → VI → VII

149. Nicotine is used for its stimulatory action. This is because it mimics the effect of
(a) testosterone (b) dopamine
(c) adrenaline (d) acetylcholine

150. The cross of superior males of one breed with superior females of another breed is referred to as
(a) outcrossing (b) cross-breeding
(c) inbreeding (d) interspecific hybridisation

151. Match the following columns and choose the correct option from the codes given below.

Column I (Term)		Column II (Definition)	
A.	Maximum natality	1.	Maximum production of individuals
B.	Ecological natality	2.	Increase in population under an actual existing specific condition
C.	Minimum mortality	3.	Minimum loss under non-limiting conditions
D.	Ecological mortality	4.	Actual loss of individuals under given environmental condition

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	1	2	4	3
(c)	1	3	2	4	(d)	3	1	2	4

152. Intake and release of O_2 from blood depends upon
(a) difference in the partial pressures of O_2 between the circulating fluid and static media
(b) CO_2 tension
(c) H^+ ion concentration
(d) All of the above

153. Blind sac body plan is not characterised by
(a) cells being organised into tissues
(b) division of labour
(c) single opening of the body for ingestion and egestion
(d) presence of two openings for digestive system

154. Ecological succession is characterised by
(a) formation and increase in thickness of soil
(b) increase in biomass
(c) higher species diversity
(d) All of the above

155. The function of DNA gyrase includes
(a) synthesis of RNA primers
(b) erases primer and fills gaps
(c) relieves torque or strain
(d) stabilises single-stranded regions

156. Identify the incorrect statement.
(a) Atrial natriuretic factor works opposite to RAAS
(b) The colour of urine is dependent upon the pigment urochrome
(c) Deamination is the removal of amine group from amino acids
(d) Nephritis is an inflammation of the urinary bladder

157. The factors responsible for Coronary Artery Disease (CAD) include
(a) hypercholesterolemia (b) hypertension
(c) diet low in antioxidants (d) All of these

158. Which among the following statements is incorrect?
(a) Luciferase enzyme responsible for light generation in fireflies is present in glyoxysomes
(b) Peroxisomes are only seen in C_3-plants where a wasteful phenomenon of photorespiration is seen
(c) Glyoxysomes are more prominent in plant seedlings
(d) Lomasomes are found to be associated with cell wall formation in plant cells

159. The CO_2 fertilisation effect relates to
(a) increased atmospheric concentration of CO_2
(b) increased productivity of C_3-plants
(c) decrease in the rate of transpiration
(d) All of the above

160. A person whose blood group is A, can have the genotype as
(a) I^AI^B, I^AI^B (b) I^AI^B, I^OI^O
(c) I^AI^A, I^AI^O (d) I^AI^O, I^BI^O

161. A graphic representation of energy flow from one trophic level to next is shaped like a pyramid because
(a) organisms at each level store most of the energy
(b) there are more producers than primary consumers
(c) eventually organisms die as they age
(d) most energy at each level is lost, leaving little for the next level

162. Identify the incorrectly matched pair.
(a) Flexor — Muscles, which bend one part of the body over the other
(b) Abductor — Muscles, which pull a limb away from the median line
(c) Depressor — Muscles, which lower some part
(d) Supinator — Muscle that turns the palm downward or backward

163. Different alleles of a gene are found on
(a) different chromosomes
(b) same chromosome
(c) homologous chromosomes
(d) same DNA strand

164. The RNA interference (RNA*i*) had been adopted to prevent the infestation of
(a) *Escherichia coli*
(b) *Meloidogyne incognita*
(c) *Bacillus thuringiensis*
(d) Lepidopterans

165. Identify the incorrectly matched pair.

(a) Palaeocene — Modernisation of angiosperms
(b) Cretaceous — Birth of modern reptiles
(c) Miocene — Mammals at height of evolution
(d) Ordovician — Land plants probably first appeared

166. Read the following statements.

I. Grave's disease is due to hypersecretion of thyroxine.
II. Cretinism is due to hyposecretion of thyroxine.
III. Acromegaly is due to hyposecretion of parathyroid.
IV. Tetany is due to hyposecretion of somatostatin.

Choose the correct option.

(a) I and II are correct (b) III and IV are correct
(c) I and II are incorrect (d) II and III are incorrect

167. Choose the incorrect statement regarding earthworm.

(a) Chloragogen cells are related to storage and synthesis of glycogen and fat
(b) Mucocytes are elongated cells with fan-like processes
(c) Phagocytes are amoeboid in nature
(d) Circular cells are enucleated

168. Which of the following is a case of recessive epistasis?

(a) Fruit colour of summer squash
(b) Coat colour of mice
(c) Flower colour in sweet pea
(d) Shape of comb in poultry

169. Cilia are present in

I. epithelium of respiratory tract
II. oviduct
III. epithelial cells of epididymis
IV. vas deferens

Choose the correct option.

(a) I and II (b) III and IV
(c) II and III (d) I and IV

170. A couple have a haemophilic son, a normal son and a haemophilic daughter. What are the most likely genotypes of the parents in this case?

Mother's genotype	Father's genotype
(a) XX^h	X^hY
(b) XX	XY
(c) X^hX^h	X^hY^h
(d) X^hY	XY^h

171. Which among the following is a major bioethical concern pertaining to genetic modification of organisms?

(a) Use of animals in biotechnology causes immense sufferings and pain to them
(b) Transfer of human genes into animals and plants weakens the concept of humanity
(c) Biotechnology may pose unforeseen risks to the environment
(d) All of the above

172. All-or-none law states that

I. the response of a muscle or nerve to a stimulus is not proportionate to intensity of stimulus.
II. the response of a muscle or nerve to a stimulus is proportionate to intensity of stimulus.
III. the degree of contraction shows independence with the intensity of stimulus.
IV. a single muscle fibre does not show any gradation in contraction in relation to degree of stimulus.

The correct statement(s) is/are

(a) I, III and IV (b) II and III (c) III and IV (d) I and III

173. Which among the following statements is/are correct regarding population density?

(a) Some newly introduced species reach high population densities
(b) Species that use abundant resources often reach higher population densities
(c) Species with small individuals generally reach higher population densities
(d) All of the above

174. Bioherbicides are

I. organisms which destroy weeds.
II. they do not harm the useful plants.
III. sporeine is bioherbicide.
IV. cochineal insect was used for the control of cacti.

The correct statement(s) is/are

(a) I and II (b) I, II and IV
(c) II and III (d) II, III and IV

175. The advantages of inbreeding include

I. it increases homozygosity.
II. it exposes harmful recessive genes that are eliminated by selection.
III. superior genes can be accumulated.

The correct statement(s) is/are

(a) I and II (b) II and III (c) I and III (d) I, II and III

176. Scotopic vision is the function of

(a) cones (b) rods (c) fovea (d) lens

177. Arrange the following in order of basic steps in genetic engineering.

I. Identification of DNA with desirable genes.
II. Maintenance of introduced DNA in host.
III. Introduction of identified DNA.

Choose the correct option.

(a) I → II → III (b) I → III → II
(c) II → I → III (d) II → III → I

178. Which component of neuron is associated with enzyme formation required for synthesis of neurotransmitters?

(a) Neurofibrils
(b) Schwann's cells
(c) Axon hillock
(d) Nissl's granules

179. In spermiogenesis,

(a) no change in number of cells as only one spermatid develops into a spermatozoa
(b) process of formation of sperms from immature germ cells
(c) number of cells increases
(d) sperms released from the seminiferous tubules

180. Which among the following statements are correct regarding archaebacteria?

I. They are obligate anaerobes.
II. Their genetic material floats freely in the cytoplasm as they do not contain nuclei.
III. They do not contain membrane bound organelles.
IV. They are found in harsh habitats such as extreme salty areas, hot springs and marshy areas.

Choose the correct option.

(a) I and II
(b) III and IV
(c) I, II and III
(d) I, II, III and IV

